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Hybrid $L_{1/2+2}$ method for gene selection in the Cox proportional hazards model

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ABSTRACT

Background and objective: An important issue in genomic research is to identify the significant genes that related to survival from tens of thousands of genes. Although Cox proportional hazards model is a conventional survival analysis method, it does not induce the gene selection.
Methods: In this paper, we extend the hybrid $L_{1/2+2}$ regularization (HLR) idea to the censored survival situation, a new edition of sparse Cox model based on the HLR method has been proposed. We develop two algorithms for solving the HLR penalized Cox model; one is the coordinate descent algorithm with HLR thresholding operator, the other is the weight iteration method.
Results: The proposed method was tested on six public mRNA data sets of serval kinds of cancers, AML, Breast cancer, Pancreatic cancer, DLBCL and Melanoma. The test results indicate that the method identified a small subset of genes but essential while giving best or equivalent predictive performance, as compared to some popular methods.
Conclusions: The results of empirical and simulations imply that the proposed strategy is highly competitive in studying high dimensional survival data among several state-of-the-art methods.
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1. Introduction

An essential issue in genomic research is to identify the significant genes that related to survival from tens of thousands of genes. The Cox proportional hazards model [1,2] is a traditional method for survival analysis, which is utilized to study the relationship between predictor features and survival time. However, because of high dimensionality and small sample size of gene-expression data, directly solving the standard Cox model is ill-posed and lead to overfitting. From the perspective of functional genomics, only a small subset of genes is related to disease development [3]. Therefore, an effective gene selection method is desired to assist in the identification of survival associated biomarkers that for clinical decision-making and prognosis.

Regularization method is one of the well-known techniques to solve the high dimensionality and small sample size problem. A typical regularization method is the least absolute shrinkage and selection operator (Lasso) penalty [4], which is often called L_1 penalty. It shrinks small coefficients to 0 and subsequently results in a sparse representation of the solution. Several L_1-type methods superior to the Lasso have been suggested in current years. Fan and Li [5] adopted the smoothly clipped absolute deviation (SCAD) penalty, which avoids immoderate penalties on big coefficients of the Lasso. Zhang and Lu [6] recommended an adaptive lasso method with an adaptively weighted penalty on coefficients. Nevertheless, the L_1-type penalizations may not be sparse enough in high-dimensional applications, and may yield inconsistent variable selections in some cases [7]. An alternative method to the L_1 method is the $L_{1/2}$ penalty [8]. The $L_{1/2}$ method enjoys various appealing properties, which includes sparsity, unbiasedness, and oracle properties [9,10]. However, just like many methods, the L-half penalty unable to deal with data with a prior structure. If there is a group of features among which the pair-wise correlations are very high, then the $L_{1/2}$ method has a tendency to pick out only one feature represents the whole group. This high correlated situation is very common in genomic data. Some studies have been proposed to tackle the problem of highly correlative genes. Engler and Li [11] suggested the penalized Cox with Elastic net penalty, in which the penalty was a linear combination of L_1 and L_2 penalties, and Becker et al. [12] proposed the Elastic SCAD (SCAD-L_2), a mix of SCAD and L_2 penalties. Such methods emphasize a group-

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ing effect, where strongly correlated genes tend to be in or out of the model together [13]. Inspired by the methods above, we proposed Hybrid $L_{1/2+2}$ Regularization (HLR) method [14], a linear combination of $L_{1/2}$ and L_2 penalties. By introducing the L_2 penalty term, HLR method also shares the grouping effect and works for the groups of predictors. The theoretical and experimental shows the HLR method can handle the high correlation data while generating a succinct solution [14].

In this paper, we propose the HLR penalized Cox model, and develop corresponding algorithms for solving the model. The model is designed to deal with the complex genomic high-dimensional data, such as microarray or next-generation datasets.

The rest of the article is structured as follows. In Part 2, we first present the Cox model and propose the HLR penalized Cox model. In Part 3, we develop two algorithms for solving the model, include a coordinate descent algorithm with HLR thresholding operator and a weight iteration method respectively. In Part 4, we evaluate the performance of our proposed method on simulation data and several real mRNA expression level datasets. A brief discussion and conclusion are presented in Part 5

2. Methods

2.1. Cox model

Suppose the dataset has n sample size with p variables and study survival time t_i on covariate X, we use the data form of $(t_1,\psi_1,x_1),\ldots,(t_n,\psi_n,x_n)$ to represent the individual's sample, where the survival time t_i being complete if $\psi_i=1$ and right censored if $\psi_i=0$. As in regression, the potential predictors are $x_i=(x_{i1},x_{i2},\ldots x_{in})$.

By using the Cox proportional hazards method, the hazard function is shown as follows:

$$h(t|\beta)=h_0(t)\exp\left(x^T\beta\right) \tag{1}$$

in which baseline hazard function $h_0(t)$ is unspecified and $\beta=(\beta_1,\beta_2,\ldots,\beta_p)^T$ represents the coefficient of p features.

With the proper dataset, the Cox partial log-likelihood [15] can be expressed as

$$l(\beta)=\sum_{i=1}^{n}\psi_i\left\{x_i^T\beta-\log\left(\sum_{j\in R_i}\exp\left(x_j^T\beta\right)\right)\right\}, \tag{2}$$

in which R_i represents the union of indices of the individuals at stake at the time t_i.

2.2. Penalized Cox model

In the view of function genomics, only of a small subset of p covariates relates to the survival. Therefore, the regularization techniques are implemented to help to identify survival pertinent features to the Cox model. The regularization approaches can pick out relevant features and estimate the regression model simultaneously by shrinking some regression coefficients to 0. By including a regularization function (or the penalty term) to Eq. (2), the penalized Cox version can be written as:

$$\hat{\beta}=\mathrm{argmin}_\beta\ \{l(\beta)+\lambda P(\beta)\}, \tag{3}$$

in which $l(\beta)$ is the loss function and $P(\beta)$ is the regularization term. A common regularization term is the L_1 (Lasso) method [4] which has the penalty function $P(\beta)=\sum_{j=1}^{p}|\beta_j|^1$. Many others L_1 type regularization functions are suggested for the Cox method in recent years, such as SCAD [5], Adaptive Lasso [6], Elastic net [11]. In the genomic study, a usual mRNA expression level dataset may reach up to 22,000 genes, and investigators desire to identify fewer but meaningful biomarkers. However, The solution of the L_1 type penalty may not be sparse enough for interpretation. Furthermore, the penalty is asymptotically biased [16,17]. Theoretically, L_0 regularization, where $P(\beta)=\sum_{j=1}^{p}|\beta_j|^0$, produce the sparsest results. However, it's going to confront with NP-hard issue. One of the alternatives to gain a better result to the Cox method is the L_q $(0<q<1)$ penalty. In which the $L_{1/2}$ penalty, with $P(\beta)=\sum_{j=1}^{p}|\beta_j|^{1/2}$, can be treated as a delegate of the L_q $(0<q<1)$ regularization and has verified many appealing characteristics, i.e., oracle, sparsity and unbiased [9]. These properties are making the $L_{1/2}$ penalty have turn out to be a powerful tool for genomic data with high dimensional [18–20]. But, the $L_{1/2}$ regularization tends to pick out one feature to symbolize the correlated union when the data are distinctly correlated. This issue may impair the result of the L-half method.

2.3. Hybrid $L_{1/2+2}$ regularization (HLR) penalized Cox model

In [14], we presented the hybrid $L_{1/2+2}$ regularization (HLR), for any fixed non-negative λ_1 and λ_2, the HLR criterion is defined as:

$$\lambda_1|\beta|_{1/2}+\lambda_2|\beta|^2 \tag{4}$$

where $\beta=(\beta_1,\ldots,\beta_p)$ are the coefficients to be estimated and

$$|\beta|_{1/2}=\sum_{j=1}^{p}\left|\beta_j\right|^{1/2},$$

$$|\beta|^2=\sum_{j=1}^{p}\left|\beta_j\right|^2.$$

Let $\alpha=\lambda_1/(1+\lambda_2)$, then Eq. (4) is equivalent to

$$\alpha|\beta|_{1/2}+(1-\alpha)|\beta|^2 \tag{5}$$

When $\alpha=0$, the HLR regularization transform into the L_2 penalty. With α set as 1, the HLR penalty turns into the L-half method. The L_2 penalty enjoys the grouping effect and the $L_{1/2}$ penalty induces sparse solutions. This combination of the two regularization methods consequently makes the HLR strategy works for the groups of predictors, and able to generate an appropriate sparse result.

Remark 1. The HLR method has strengthened the power of prediction accuracy of $L_{1/2}$ regularization by a particular mechanism. More details can be referred to [14].

The theoretical and experimental test demonstrates that the HLR regularization is a robust method for high dimensional data. Thus, the HLR penalized Cox model is naturally expected:

$$\beta_{HLR}=\mathrm{argmin}_\beta\ \left\{l(\beta)+\lambda\left(\alpha|\beta|_{1/2}+(1-\alpha)|\beta|^2\right)\right\} \tag{6}$$

3. Algorithms for HLR penalized Cox model

For further release the effectiveness of the proposed method, in this section, we present two algorithms, include an efficient HLR thresholding operator method and a weight iteration approach respectively, for solving HLR penalized Cox model. To begin with the solutions, we first need to transform the nonlinear Cox model into a linear model. Here, we follow the methods from Tibshirani [4] and Gui and Li [21] to convert the Cox partial log-likelihood function (Eq. (2)) into a linear regression problem.

Let $\eta=\beta^Tx$, $\zeta=x^T\beta$, $k=-\frac{\partial l}{\partial\zeta}$, $V=-\frac{\partial^2 l}{\partial\zeta\partial\zeta^T}$, and $z=\zeta+V^-k$, where V^- represents an inverse to V. By using the Taylor expansion of function $l(\beta)$ in Eq. (2), the partial log-likelihood can be rewritten as:

$$l(\beta)\approx(z-\zeta)^TV(z-\zeta) \tag{7}$$

We use the Cholesky decomposition to obtain $Q = V^{1/2}$ such that $Q^T Q = V$, $\hat{y} = Qz$ and $\hat{x} = Qx$, thus, the Eq. (7) can be expressed as:

$$l(\beta) \approx \left(\hat{y} - \hat{x}^T\beta\right)^T\left(\hat{y} - \hat{x}^T\beta\right). \tag{8}$$

Then, the HLR penalized Cox model can be defined by

$$\beta_{HLR} = \text{argmin}_\beta \left\{\left(\hat{y} - \hat{x}^T\beta\right)^T\left(\hat{y} - \hat{x}^T\beta\right) + \lambda\left(\alpha|\beta|_{1/2} + (1-\alpha)|\beta|^2\right)\right\}. \tag{9))}$$

3.1. A coordinated descent algorithm for HLR penalized Cox model by using HLR thresholding operator method

Let dataset X have n observations $X = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$, in which $x_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ represents ith observation with p dimensional, and y_i represents the dependent variable. The decision matrix is normalized as: $\sum_{i=1}^{n} x_{ij}^2 = 1$.

The coordinate descent (CD) algorithm [22] is a powerful technique for deal with regularization model especially with high dimensional data, because its asymptotic time complexity is only O(mnp), in which m is the number of iterations, and the values of m and n are normally small.

In [14], we presented an efficient coordinate-wise update algorithm for the HLR method:

$$\beta_j \leftarrow \frac{Half\left(\omega_j,\ \lambda\alpha\right)}{1 + \lambda(1-\alpha)} \tag{10}$$

where $\omega_j = \sum_{i=1}^{n} x_{ij}(y_i - \tilde{y}_i^{(j)})$, and $\tilde{y}_i^{(j)} = \sum_{k \neq j} x_{ik}\beta_k$ as the partial residual for fitting β_j. *Half*(ʒ, τ) is the $L_{1/2}$ thresholding operator:

$$Half\left(\omega_j, \lambda\right) = \begin{cases} \frac{2}{3}\omega_j\left(1 + \cos\left(\frac{2(\pi - \varphi_\lambda(\omega_j))}{3}\right)\right) & if\ \left|\omega_j\right| > \frac{\sqrt[3]{54}}{4}(\lambda)^{\frac{2}{3}} \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

where $\varphi_\lambda(\omega) = arccos(\frac{\lambda}{8}(\frac{|\omega|}{3})^{-\frac{3}{2}})$, $\pi = 3.14$.

The convergence of $L_{1/2}$ thresholding operator algorithm has been proved in [10], which implies that the HLR coordinate-wise update algorithm (Eq. (10)) will reach to the sets of minima (global or local) of the problems.

The CD algorithm with the HLR coordinate-wise update operator to the sparse Cox model is (Algorithm 1):

3.2. A weight iteration algorithm for HLR penalized Cox model

The HLR method can be transformed into a weighted version of the Elastic net, which the standard Elastic net solutions can be directly employed. The weight iteration algorithm is shown as follows (Algorithm 2):

In the above solution, we have used the value of E as the termination condition. In Step 1, the β^0 can be initialized as $\beta^0 = (1, 1, \ldots, 1)$. Interestingly, with such an initialization, when $m = 0$ in Step 2 is precisely leading to an Elastic net result. When in the second iteration ($m = 1$), Step 2 is to deal with a weight Elastic net problem, which can be converted into an Elastic net regularizer by means of linear transformation. Thus, the result yielded by the Algorithm 2 must be more ideal than those of Elastic net. There is possible that when m $\geq$ 1, some β_i are 0. Thus, to maintain the feasibly, in Step 2, $\frac{1}{\sqrt{|\beta_i^m|}}$ can be replaced by $\frac{1}{\sqrt{|\beta_i^m|+\varepsilon}}$ when implementing, which ε is a positive real number.

We then analyze the convergence of the Algorithm 2 and have the subsequent theorem.

Theorem 1. The Algorithm 2 converges to the global or local minima when $m \to \infty$.

Proof. ***of Theorem 1:***

Let $G_n(\beta) = L_n(\beta) + \lambda(a\sum_{i=1}^{p}\frac{1}{\sqrt{|\beta_i^m|}}|\beta_i| + (1-a)\sum_{i=1}^{p}|\beta_i|^2)$ and

$$G_n^*(\beta) = L_n(\beta) + \lambda\left(a\sum_{i=1}^{p}\sqrt{|\beta_i|} + (1-a)\sum_{i=1}^{p}|\beta_i|^2\right).$$

The Eq. (9) can be rewritten as

$$\beta_{HLR} = argmin_{\beta^+,\beta^-}\left\{\frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}_i - \hat{x}_i^T\left(\beta^+ - \beta^-\right)\right)^2 + \lambda\left(a\sum_{i=1}^{p}\sqrt{\beta^+{}_i + \beta^-{}_i} + (1-a)\sum_{i=1}^{p}\left(\beta^+{}_i + \beta^-{}_i\right)^2\right)\right\},$$

where positive part $\beta^+ = \max(\beta, 0)$ and negative part $\beta^- =$-min (β, 0). Then, we have $\beta = \beta^+ - \beta^-$ and $|\beta| = \beta^+ + \beta^-$. And we get

$$G_n(\beta) = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}_i - \hat{x}_i^T\left(\beta^+ - \beta^-\right)\right)^2 + \lambda\left(a\sum_{i=1}^{p}\frac{1}{\sqrt{\beta^{+m}{}_i + \beta^{-m}{}_i}}\left(\beta^+{}_i + \beta^-{}_i\right) + (1-a)\sum_{i=1}^{p}\left(\beta^+{}_i + \beta^-{}_i\right)^2\right).$$

Let $\beta^m = (\beta^+,\ \beta^-)^m$, and thus Step 2 in the Algorithm 2 is exactly to solving $G_n(\beta)$; Thus β^{m+1} meets

$$\vec{\nabla}L_n\left(\beta^{m+1}\right) = -\lambda\vec{\nabla}P(\beta^m). \tag{13}$$

Because of the $L_n(\beta)$ is convex, which suggests

$$L_n(\beta^m) \geq L_n\left(\beta^{m+1}\right) + \left(\beta^m - \beta^{m+1}\right)\vec{\nabla}L_n\left(\beta^{m+1}\right), \tag{14}$$

for all β^m, β^{m+1}. While $\lambda(a\sum_{i=1}^{p}\sqrt{\beta^+{}_i + \beta^-{}_i} + (1-a)\sum_{i=1}^{p}(\beta^+{}_i + \beta^-{}_i)^2)$ normally is a concave function about β^+, β^-. For all β^m and β^{m+1}, we have

$$\lambda P\left(\beta^{m+1}\right) \leq \lambda P(\beta^m) + \left(\beta^{m+1} - \beta^m\right)\lambda\vec{\nabla}P(\beta^m) \tag{15}$$

With Eq. (14), (15) and using (13), we have

$$G_n^*\left(\beta^{m+1}\right) \leq G_n^*(\beta^m).$$

Therefore, G_n^* is a function which monotonically decreasing and bounded. As $m \to \infty$, β^m converges to the set of stationary points of $G_n^*(\beta)$ by the Lasalle invariance principle.

Theorem 1. dedicates that the Algorithm 2 will consistently approach to the union of minimizers of $G_n^*(\beta)$ with local or global, and $G_n^*(\beta^m)$ will reach to its minima (local or global).

Remark 2. There are many solutions to solve the Elastic net. For example, the CD algorithm [22]. The key factor in designing a CD algorithm is the thresholding operator of the penalty function. Following Friedman et al. [22], we induce a Lasso type thresholding operator to the Equation (12), that is

$$\beta_j^{m+1} \leftarrow \frac{S\left(\omega_j,\ \frac{\lambda\alpha}{\sqrt{|\beta_j^m|}}\right)}{1 + \lambda(1-\alpha)},$$

Algorithm 1

Step 1: Set $\beta_j(m) \leftarrow 0$ ($j=1, 2, \ldots, p$) and $m \leftarrow 0$, Initialize x, y, λ and α;
Step 2: Calculate $\zeta(m)$, $k(m)$, $V(m)$, $\hat{x}(m)$ and $\hat{y}(m)$ based on the current $\beta(m)$;
Step 3: Calculate $(\hat{y}-\hat{x}^T\beta)^T(\hat{y}-\hat{x}^T\beta)+\lambda(\alpha|\beta|_{1/2}+(1-\alpha)|\beta|^2)$ as repeat following:
Renew $\beta_j(m)$, and cycle over j=1,..., p:
Set $\tilde{y}_i^{(j)}(m) \leftarrow \sum_{k\neq j}\hat{x}_{ik}\beta_k(m)$
and $\omega_j(m) \leftarrow \sum_{i=1}^{n}\hat{x}_{ij}(\hat{y}_i(m)-\tilde{y}_i^{(j)}(m))$;
Update $\beta_j(m) \leftarrow \frac{Half(\omega_j(m),\lambda\alpha)}{1+\lambda(1-\alpha)}$;
Step 4: Let $m \leftarrow m+1$, repeat Steps 2 and 3 until $\sum_{i=1}^{p}|\beta_i(m)-\beta_i(m-1)| < 10^{-5}$;

where S(z, γ) is the soft-thresholding operator with value

$$\text{sign}(z)(|z|-\gamma)+=\begin{cases} z-\gamma \ if\ z>0\ and\ \gamma<|z| \\ z+\gamma \ if\ z<0\ and\ \gamma<|z| \\ z-\gamma \ if\ \gamma\geq|z|. \end{cases}$$

Then, the CD method for the Algorithm 2 is ready, and its procedure is similar to those of the Algorithm 1 except the thresholding operator update rule. Its asymptotic time complexity also is O(mnp).

The optimal regularization parameters are tuned by the 10-fold cross-validation (CV) [23] approach in training set on two-dimensional (λ and α for Algorithm 1) or three-dimensional (λ, α and ε for Algorithm 2) parameter surfaces.

4. Results and discussion

4.1. Landmark measures

In this paper, we use two validation methods for prediction performance measurement, include the integrated Brier score (IBS) [24] and the concordance index (CI) [25] respectively:

4.1.1. Integrated brier-score (IBS)

The BS depends on time t. As a result, it is reasonable to apply the IBS as an index to measure the goodness of the predicted survival functions of all samples between time zero and an arbitrary upper limit t^*.

The Brier Score (BS) can be described as:

$$BS(t)=\frac{1}{n}\sum_{i=1}^{n}\left[\frac{\hat{S}(t|x_i)^2 1(t_i\leq t\wedge\psi_i=1)}{\hat{G}(t_i)}+\frac{\left(1-\hat{S}(t|x_i)^2\right)1(t_i>t)}{\hat{G}(t)}\right], \tag{16}$$

in which $t>0$, $\hat{G}(\cdot)$ represents the Kaplan-Meier estimation of the censoring distribution and $\hat{S}(\cdot|X_i)$ stands to estimate survival for patient i. Note that the $BS(t)$ is dependent on the point at time t, and its values are between 0 and 1. The correct predictions at time t result in small values of BS. The IBS is given by:

$$IBS=\frac{1}{\max(t_i)}\int_0^{\max(t_i)}BS(t)dt. \tag{17}$$

4.1.2. Concordance index (CI)

The CI can be treated as the ratio of the pairs of observations which the survival times ordered are correctly predicted by the model among the observation that ordered. With that definition, we have $f_i<f_j$ while $t_i<t_j$ and $\psi_i=1$ in which $f(\bullet)$is the survival function.

Therefore, the CI criterion can be expressed as:

$$CI=\frac{\sum_i\sum_j 1\left(f_i<f_j\wedge\psi_i=1\right)}{\sum_i\sum_j 1\left(t_i<t_j\wedge\psi_i=1\right)}. \tag{18}$$

Note that when CI equals to 1, implies a perfect prediction ability of the model, when CI <= 0.5 indicates at random.

4.2. Simulation

In this section, we perform a simulation research to measure the feature selection, and prediction capacity of the HLR sparse Cox model. Five sparse Cox model techniques: Lasso, Elastic net, $L_{1/2}$, SCAD and SCAD-L_2 are compared in the experiment. Simulation scheme is similar to Bender's work [26].

Detailed procedures for survival data generation are:

i. We first create a decision matrix $X=\{x_1,\ldots,x_n\}\sim N(0,\ 1)$, in which, n is observation size.
ii. The survival time T_i (i=1,..., n) is generated from a uniformly distributed variable U by $T_i=(1/\theta)log(1-(\theta log(U)/\omega exp(x_i\beta)))$, where ω is the scale variable, θ represents the shape variable, β is the ground-true regression coefficients.
iii. Censoring time point T_i' (i=1,...,n) is generated from an exponential distribution $E(\alpha)$, where α is determined by specify censoring rate.
iv. Define $t_i=\min(T_i,\ T_i')$ and $\psi_i= I(T_i\leq T_i')$, then, the test data represented as (t_i, ψ_i, x_i) for the Cox model (Eq. (2)) are generated.

We provide four simulation settings in this paper. For every setting, the dimension of predictor p is set as 1100. The symbol **./.** is the number of samples in the training and test sets respectively, e.g., 200/200. The details of the four settings are as follows:

(a) In setting 1, the dataset includes 100/100 samples and five prognostic genes

$$\beta\left(\underbrace{1,\ 1,1,\ 1,\ 1,}_{5}\quad\underbrace{0,\ldots,\ 0}_{1195}\right).$$

we simulated a grouped feature case:

$$x_i=\rho\times x_1+(1-\rho)\times x_i,\ \text{i}=2,3,4,5.$$

where ρ is control variable of the grouped features.

(b) The setting 2 is similar to the setting 1, but we consider the situation while exiting other independent prognostic genes relate to survival,

$$\beta\left(\underbrace{1,1,1,\ 1,\ 1,\ 0.2,\ -0.2,\ 0.7,\ 3,\ -1.5,}_{10}\quad\underbrace{0,\ldots,\ 0}_{1090}\right)$$

(c) In setting 3, the dataset consists of 200/200 observations and two grouped features,

$$\beta\left(\underbrace{1,1,1,\ 1,\ 1,\ 0.2,-0.2,\ 0.7,\ 3,\ -1.5,}_{10}\quad\underbrace{2,\ldots,2,}_{10}\quad\underbrace{0,\ldots,\ 0}_{1070}\right)$$

$$x_i=\rho\times x_1+(1-\rho)\times x_i,\ \text{i}=2,3,4,5.$$

$$x_i=\rho\times x_{11}+(1-\rho)\times x_i,\ \text{i}=12,\ldots,30.$$

(d) In setting 4, the prognostic genes are increased to about 18% of the whole genes, and the dataset includes 400/400 sample size, and with

$$\beta=\left(\underbrace{2,\ldots,2,}_{15}\quad\underbrace{-1.5,\ 1,-1.5,\ 2,-2.5,}_{5}\quad\underbrace{4,\ldots,\ 4,}_{20}\right.$$

Algorithm 2

Step 1: Initialize value β^0 and the terminal variable E. Let m=zero.
Step 2: Calculate
$$\beta^{m+1} = \text{argmin}\{\sum_{i=1}^{n}(\hat{y}_i - \hat{x}_i^T\beta)^T(\hat{y}_i - \hat{x}_i^T\beta) + \lambda(a\sum_{i=1}^{p}\frac{1}{\sqrt{|\beta_i^m|}}|\beta_i| + (1-a)\sum_{i=1}^{p}|\beta_i|^2)\} \quad (12)$$
with a standard Elastic net solution, and let $m := m+1$.
Step 3: If $E > m$, repeat Step 2, else output the result β^m.

$$\underbrace{1,\ldots,1,}_{45}\quad \underbrace{2,\ldots,2,}_{45}\quad \underbrace{3,\ldots,3,}_{70}\quad \underbrace{0,\ldots,0}_{900}\Bigg)$$

we defined three grouped variables

$$x_i = \rho \times x_1 + (1-\rho)\times x_i,\ \mathrm{i} = 2,\ldots,30;$$

$$x_i = \rho \times x_{36} + (1-\rho)\times x_i, \mathrm{i} = 37,\ldots,75.$$

$$x_i = \rho \times x_{101} + (1-\rho)\times x_i, \mathrm{i} = 102,\ldots,130.$$

In those settings, a perfect penalized regression approach would pick out only real prognostic genes, and assign the coefficients of the noise variables to 0.

In the simulation, about 30% of the data are right censored. We set the correlation control variable ρ of genes to 0.2, 0.5 and 0.9 severally. We replicated the experiment 600 times for every regularization approach, and assessed the average prediction accuracy (IBS and CI) on the test set. For measuring the performance of the identified genes in training set; the sensitivity and specificity of the gene selection [27] were used and can be shown as follows:

$$\text{True Negative (TN)} := \left|\bar{\beta}.*\overline{\hat{\beta}}\right|_0,\quad \text{False Positive (FP)} := \left|\bar{\beta}.*\hat{\beta}\right|_0$$

$$\text{False Negative (FN)} := \left|\beta.*\overline{\hat{\beta}}\right|_0,\ \text{True Positive (TP)} := \left|\beta.*\hat{\beta}\right|_0$$

$$\text{Sensitivity} := \frac{\text{TP}}{\text{TP}+\text{FN}},\quad \text{Specificity} := \frac{\text{TN}}{\text{TN}+\text{FP}}.$$

where the .* is the element-wise product, and $|.|_0$ counts the non-zero entity in a vector, $\bar{\beta}$ and $\overline{\hat{\beta}}$ are the "not" logical operators on the $\hat{\beta}$ and β.

Table 1 shows the average performance of gene section by several regularization methods for the Cox model in 600 runs. In general, the Elastic net records the lowest specificity indicates the approach identified the biggest number of noise features. The HLR technique performs very competitively regarding specificity meanwhile result in the highest sensitivity than the Lasso, SCAD, Elastic net, L-half and SCAD-L_2 approaches, which implies that the HLR method selected fewer noise genes and the largest number of real prognostic genes. For example, in the setting 2 with $\rho = 0.5$, The HLR method (with Algorithm 1) achieves the superior performance, with successfully select the whole 10 true genes (sensitivity = 1), and a few noise genes (specificity = 0.997).

Table 2 shows the average prediction performance (IBS and CI) for the several methods in 600 runs. In general, the HLR approach seems to get a little advantage over than the other methods regarding IBS and CI measurement. For example, when $\rho = 0.2$ and in the setting 2, the mean of IBS and CI from the HLR method (with Algorithm 1) are 0.079 and 0.914, which are the best performance amongst all the competitors. When $\rho = 0.5$ and in the setting 2, the mean of IBS from the HLR method (with Algorithm 2) is 0.080, better than 0.088, 0.082, 0.096, 0.095 and 0.089 get by Lasso, SCAD, Elastic net, L-half and SCAD-L_2 penalties. When $\rho = 0.9$ and in setting 2, the mean of CI from the HLR method (with Algorithm 2) is 0.790, better than 0.780, 0.712, 0.726, 0.709 and 0.775 get by the other methods. Combined with the outcomes showed in Table 1, we concluded that the HLR penalized approach demonstrated in a superior way or comparable predictive performance and with a better accurate model, especially when the data are highly correlated, to the other strategies.

We also performed a convergence test for the methods, and the convergence time amongst the algorithms was similar. For example, in the setting 2 with $\rho = 0.5$, it took MATLAB 30, 24, 32, 45, 26, 29 and 33 s to converge for the Lasso, SCAD, Elastic net, L-half , SCAD-L_2, HLR-1 (with Algorithm 1) and HLR-2 (with Algorithm 2) methods.

4.3. Analyzes of gene expression data

For further estimate our method, we employed six publicly available datasets for measurement. We added two traditional approaches for a comprehensive comparison: principal components regression (PCR) [28] and partial least squares (PLS) [29].

Compact information of the six datasets is given as follows and sums up in Table 3.

AML Dataset [30]. It includes 116 AML patients in which the censored cases are 49, and each patient contains 6283 genes.

Breast Cancer Dataset [31]. It includes 107 breast cancer samples, and each sample contains 21,025 genes.

Pancreatic Cancer Dataset [32]**.** It contains 63 pancreatic cancer patients, and each patient contains 22,115 genes.

Diffuse Large B-cell Lymphoma Dataset (DLBCL) [33]. It includes 240 DLBCL patients in which the censored samples are 102, and each patient contains 7399 genes.

Melanoma Dataset [34]. It consists of 210 Melanoma samples, and every case includes 21,592 genes.

Lung Cancer Dataset [35]. It includes 67 lung cancer patients in which the censored samples are 49, and each patient consists of 19,222 genes.

To measure the efficacy of the HLR sparse Cox model, we separate the datasets randomly, so that approximate 2/3 of the observations turn into the training samples, and the rest 1/3 of the observations as the test cases for validation. For estimating optimal tuning parameters, we employed the ten-fold CV in the training data. The above procedures are repeated 100 times. The average number of the selected genes, and testing IBS and CI, are summarized in Tables 4 and 5.

Table 4 displays the mean number of genes selected by all the methods. In general, the HLR regularization method performs better than PCR, PLS, Lasso, SCAD, Elastic net and SCAD-L_2, and similar to $L_{1/2}$. For example, in the Pancreatic dataset, the HLR penalized method (with Algorithm 1) selected about 5.4 genes, compared to about 22,115, 22,115, 13.1, 7.5, 11, 5.6 and 21.5 selected by PCR, PLS, Lasso, SCAD, Elastic net, $L_{1/2}$ and SCAD-L_2. For Lung cancer dataset, the best one is the $L_{1/2}$ penalty and the second is HLR method (with Algorithm 1).

As demonstrated in Table 5, in general, the HLR penalized method outperforms than the other penalized methods, although, the performances are not significantly different among all regularization methods. For example, for AML dataset, the HLR method (with Algorithm 1) gains the mean ten-fold CV test IBS of 0.145, and the mean CI of 0.819 with approximately 18.7 genes selected. The models with PCR, PLS, Lasso, SCAD, Elastic net, $L_{1/2}$ and SCAD-

Table 1
Average outcomes of the simulated experiment.

ρ	Method	Scenario							
		1	2	3	4	1	2	3	4
		Sensitivity of feature selection				Specificity of feature selection			
	Lasso	1.000	0.933	0.867	0.655	0.987	0.953	0.906	0.683
	SCAD	1.000	0.938	0.823	0.607	0.998	0.971	0.930	0.787
0.2	EN	1.000	**0.967**	0.900	0.812	0.971	0.895	0.887	0.622
	$L_{1/2}$	1.000	0.937	0.622	0.415	0.998	0.980	**0.931**	0.819
	SCAD-L_2	1.000	0.961	0.908	**0.819**	0.965	0.890	0.865	0.737
	HLR-1	1.000	0.950	**0.911**	0.790	0.997	0.950	0.882	**0.834**
	HLR-2	1.000	0.933	0.889	0.801	**1.000**	**0.991**	0.869	0.781
	Lasso	1.000	0.900	0.633	0.385	0.999	0.975	0.862	0.882
	SCAD	1.000	0.886	0.629	0.409	1.000	0.990	0.833	0.919
	EN	1.000	0.967	0.856	0.643	1.000	0.898	0.942	0.839
0.5	$L_{1/2}$	0.933	0.700	0.267	0.203	0.979	0.978	0.926	0.896
	SCAD-L_2	0.979	0.978	0.926	0.696	1.000	0.947	**0.969**	0.879
	HLR-1	1.000	**1.000**	0.851	0.707	0.986	**0.997**	0.967	**0.922**
	HLR-2	1.000	1.000	0.856	**0.718**	1.000	0.994	0.879	0.871
	Lasso	0.933	0.700	0.278	0.167	0.863	0.936	0.972	0.940
	SCAD	0.926	0.711	0.325	0.151	1.000	0.955	0.989	0.962
	EN	1.000	0.933	0.867	0.607	1.000	0.931	0.954	0.899
0.9	$L_{1/2}$	0.900	0.693	0.122	0.108	1.000	**0.998**	**0.993**	0.955
	SCAD-L_2	1.000	0.861	0.809	0.617	1.000	0.926	0.944	0.881
	HLR-1	1.000	0.967	0.860	0.641	1.000	0.976	0.977	**0.968**
	HLR-2	1.000	**1.000**	0.867	**0.643**	1.000	0.967	0.950	0.819

The outcomes are based on 600 times. The sensitivity and specificity are both aimed to estimate the performance of the identified features. EN represents the Elastic net method. HLR-1 and HLR-2 represent the HLR approach with algorithms 1 and 2, respectively. In bold – the best performance among all the methods.

Table 2
Mean IBS and CI results by the six regularization approaches on the simulated data in 600 runs.

ρ	Method	Scenario							
		1	4	2	3	1	2	3	4
		IBS				CI			
	Lasso	0.090	0.080	0.095	0.095	0.827	0.878	0.777	0.836
	SCAD	0.088	0.081	**0.089**	0.095	0.831	0.889	0.773	0.858
0.2	EN	0.096	0.082	0.096	**0.088**	**0.834**	0.898	0.769	0.839
	$L_{1/2}$	**0.075**	0.089	0.090	0.095	0.797	0.892	**0.817**	0.877
	SCAD-L_2	0.083	0.080	0.092	0.092	0.822	0.855	0.732	0.803
	HLR-1	0.106	**0.079**	0.090	0.089	0.803	**0.914**	0.814	**0.878**
	HLR-2	0.077	0.080	0.106	0.091	0.785	0.878	0.781	0.822
	Lasso	0.085	0.088	0.098	0.092	0.765	0.739	0.741	0.759
	SCAD	0.057	0.082	0.086	0.090	0.777	0.841	**0.765**	0.788
0.5	EN	0.088	0.096	0.088	0.092	0.764	**0.846**	0.724	0.755
	$L_{1/2}$	0.075	0.095	**0.071**	0.100	0.784	0.838	0.741	**0.793**
	SCAD-L_2	0.077	0.089	0.091	0.096	0.764	0.822	0.740	0.758
	HLR-1	**0.049**	0.082	0.099	**0.090**	**0.785**	0.777	0.717	0.788
	HLR-2	0.054	**0.080**	0.077	0.093	0.755	0.786	0.691	0.750
	Lasso	**0.060**	0.092	0.090	0.077	**0.794**	0.780	0.696	0.736
	SCAD	0.064	0.089	0.090	0.077	0.791	0.712	0.699	0.752
0.9	EN	0.084	**0.089**	0.091	0.077	0.694	0.726	0.685	0.745
	$L_{1/2}$	0.097	0.090	0.091	0.077	0.696	0.709	0.695	**0.758**
	SCAD-L_2	0.091	0.092	0.090	0.077	0.714	0.775	0.693	0.732
	HLR-1	0.083	0.095	0.090	0.077	0.718	0.763	**0.700**	0.753
	HLR-2	0.084	0.092	0.093	0.078	0.716	**0.790**	0.689	0.745

Table 3
mRNA expression level datasets used throughout this article.

Dataset	No. of samples (Total)	No. of censored	No. of genes
AML	116	49	6283
Breast cancer	107	78	21,025
Pancreatic cancer	63	21	22,115
DLBCL	240	102	7399
Melanoma	210	108	21,592
Lung cancer	67	49	19,222

L_2 methods record the mean ten-fold CV to IBS of 0.187, 0.193, 0.198, 0.203, 0.192, 0.197 and 0.202, and the average CI of 0.801, 0.789,0.761, 0.736, 0.812, 0.800, 0.770 with 6283, 6283, 35.2, 15.8, 41.6, 13.1 and 30.5 genes selected respectively, in which the HLR Cox model (with Algorithm 1) is slightly better than the other methods.

For Melanoma datasets, the HLR method (with Algorithm 1) gains the maximum IBS value. For lung cancer, the testing performance of CI of our method (with Algorithm 1) is the same as Elastic net; however, the HLR only using 13.3 genes to gains its achievement, compared to 21.6 genes for the Elastic net method. Even though the SCAD-L_2 method earned the best performance of IBS in lung cancer, the CI performance and the number of gene selection of the approach are no match for the HLR technique. In the

Table 4
The mean number of the gene selected by the nine approaches on the six mRNA datasets.

Datasets	PCR	PLS	Lasso	SCAD	EN	$L_{1/2}$	SCAD-L_2	HLR-1	HLR-2
AML	6283	6283	35.2	15.8	41.6	**13.1**	30.5	18.7	22.3
Breast cancer	21,025	21,025	21.9	23.1	38.3	**17.2**	26.3	23.5	22.7
Pancreatic cancer	22,115	22,115	13.1	7.5	11	5.6	21.5	**5.4**	7.4
DLBCL	7399	7399	7	8	11.7	7.6	9.3	**6**	7.7
Melanoma	21,592	21,592	25	30.2	43	20.2	15.2	**12.6**	17.9
Lung cancer	19,222	19,222	14	13.5	21.6	**9.2**	14.2	13.3	13.7

Table 5
The IBS and CI results achieved by the several approaches on the six mRNA datasets.

Datasets	Average IBS								
	PCR	PLS	Lasso	SCAD	EN	$L_{1/2}$	SCAD-L_2	HLR-1	HLR-2
AML	0.187	0.193	0.198	0.203	0.192	0.197	0.202	**0.145**	0.193
Breast cancer	0.151	0.154	0.156	0.179	0.156	0.168	0.156	**0.148**	0.151
Pancreatic cancer	0.150	0.149	0.157	**0.148**	0.157	0.184	0.162	0.188	0.165
DLBCL	0.197	0.221	0.200	0.208	0.206	0.218	**0.197**	0.207	0.207
Melanoma	0.188	**0.183**	0.195	0.197	0.193	0.185	0.188	0.187	0.191
Lung cancer	0.172	0.184	0.195	0.185	0.184	0.182	**0.162**	0.170	0.211
Datasets	Average CI								
	PCR	PLS	Lasso	SCAD	EN	$L_{1/2}$	SCAD-L_2	HLR-1	HLR-2
AML	0.801	0.789	0.761	0.736	0.812	0.800	0.770	**0.819**	0.805
Breast cancer	0.793	0.781	0.768	0.772	0.775	0.818	0.794	0.807	**0.819**
Pancreatic cancer	0.822	0.819	0.772	**0.830**	0.804	0.797	0.785	0.827	0.814
DLBCL	**0.783**	0.772	0.761	0.777	0.757	0.768	0.762	0.776	0.751
Melanoma	0.835	0.841	0.811	0.784	0.841	0.796	0.817	**0.846**	0.832
Lung cancer	0.847	0.839	0.813	0.818	**0.849**	0.847	0.828	**0.849**	0.831

Table 6
Genes of the most frequently 10 biomarkers identified by the HLR approach from the breast cancer dataset.

Gene name	Gene description
MCPH1	Microcephalin 1
ARMC7	Armadillo repeat containing 7
ACADM	Acyl-CoA dehydrogenase, C-4 to C-12 straight chain
ARRB1	Arrestin, beta 1
WNT3	Wingless-type MMTV integration site family, member 3
KANK3	KN motif and ankyrin repeat domains 3
ERBB2	Erb-b2 receptor tyrosine kinase 2
DNAJB6	DnaJ (Hsp40) homolog, subfamily B, member 6
TRIM16	Tripartite motif containing 16
MLC1	Megalencephalic leukoencephalopathy with subcortical cysts 1

survival analysis, where the objective is often to develop an accurate prediction model using as few genes as desired in orders to control cost, and the HLR method is promising to the objective.

4.4. Biomarker identification using HLR method in human breast cancers

We demonstrated the most frequently 10 biomarkers identified by the HLR penalized method (with Algorithm 1) from the breast cancer dataset in Table 6. Note that in simulation experiment, the approach has shown its capacity to identified the true prognostic genes. It was implied that the features identified by the HLR approach from the six disease datasets might also beneficial to the investigators whose goal is to look for the key prognostic biomarkers that related to the tumour treatment. For instance, as shown in Table 6, the prognostic genes selected by our HLR method include MCPH1, also known as BRIT1, has recently been identified as a novel key regulatory gene of the DNA damage response pathway. As such, MCPH1 is aberrantly expressed in many malignancies, especially in breast cancer, and its function has been implicated in breast cancer suppression [36]. The estrogen receptor (ESR1), is one of the essential genes to be used for clinically categorized Breast cancer, this biomarker is routinely used not only to define prognosis but also determine treatment [37]. DNAJB6, is a novel member of the human DnaJ family, has been shown plays a critical role in breast cancer. DNAJB6 is a nuclear protein that is lost in breast cancer, which regulates several key players in tumor formation and metastasis, and that is functionally able to retard tumor growth [38].

Consideration of the outcomes showed in Tables 4, 5 and 6; we observed that the HLR penalized approach identified a small subset of genes but essential while giving best or equivalent predictive performance.

5. Discussion and conclusion

In our pervious study [14], we presented the HLR method, and applied the method to the logistic regression model for cancer classification, the results from [14] are promising. For better identify the significant genes that related to survival from tens of thousands of genes. In this paper, we extended the HLR theory to the censored survival situation, a new edition of sparse Cox model based on the HLR method has been proposed. We developed two novel algorithms for solving the HLR penalized Cox model; one was the coordinate descent algorithm with HLR thresholding operator, the other was the weight iteration method. We have proven the algorithms will consistently approach to the union of minimizers with local or global. A more thoughtful analysis and extensive experiments were present in this paper compare with [14].

One of the central goals of survival analysis is to identify a few of main risk features. In this paper, we proposed the HLR penalize Cox method for survival analysis. The suggested method is appli-

cable to the complex genomic data with the characteristic of large p and small n. The practical result and simulations show the HLR penalized Cox model is highly competitive in analyzing high dimensional survival data, because the method can induce stronger sparsity and better correctness. The proposed method is a promising implement for survival analysis, and might be useful for the search for new cancer targets for targeted therapy.

Conflicts of interest

None.

Authors' contributions

Hai-Hui Huang developed the method, performed the analysis and drafted the article. Yong-Liang initiated, and supervised the project. All authors have read the final version of the article and have approved submission of the manuscript to Computer Methods and Programs in Biomedicine.

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References

[1] M. Melisko, A multigene assay to predict recurrence of tamoxifen-treated, node-negative breast cancer, Women's Oncol. Rev. 5 (2005) 45–47, doi:10.3109/14733400500093379.
[2] Fibrinogen Studies Collaboration, Measures to assess the prognostic ability of the stratified Cox proportional hazards model, Stat. Med. 28 (2009) 389–411, doi:10.1002/sim.3378.
[3] K.R. Schultz, D.B. Miklos, D. Fowler, K. Cooke, J. Shizuru, E. Zorn, E. Holler, J. Ferrara, H. Shulman, S.J. Lee, P. Martin, A.H. Filipovich, M.E.D. Flowers, D. Weisdorf, D. Couriel, P.A. Lachenbruch, B. Mittleman, G.B. Vogelsang, S.Z. Pavletic, Toward biomarkers for chronic graft-versus-host disease: national institutes of health consensus development project on criteria for clinical trials in chronic graft-versus-host disease: III. Biomarker working group report, Biol. Blood Marrow Transplant. 12 (2006) 126–137, doi:10.1016/j.bbmt.2005.11.010.
[4] R. Tibshirani, The Lasso method for variable selection in the cox model, Stat. Med. 16 (1997) 385–395. doi: 10.1002/(SICI)1097-0258(19970228)16:4(385::AID-SIM380)3.0.CO;2-3.
[5] J. Fan, R. Li, Variable selection for cox ' s proportional hazards model and frailty model, Ann. Stat. 30 (2002) 74–99. http://www.jstor.org/stable/2700003. (Accessed 31 May 2017).
[6] H.H. Zhang, W. Lu, Adaptive Lasso for Cox's proportional hazards model, Biometrika 94 (2007) 691–703, doi:10.1093/biomet/asm037.
[7] H. Zou, The adaptive lasso and its oracle properties, J. Am. Stat. Assoc. 101 (2006) 1418–1429.
[8] C. Liu, Y. Liang, X.-Z. Luan, K.-S. Leung, T.-M. Chan, Z.-B. Xu, H. Zhang, The L1/2 regularization method for variable selection in the Cox model, Appl. Soft Comput. 14 (2014) 498–503, doi:10.1016/j.asoc.2013.09.006.
[9] Z. Xu, H. Zhang, Y. Wang, X. Chang, Y. Liang, L1/2 regularization, Sci. China Inf. Sci. 53 (2010) 1159–1169.
[10] Z. Xu, X. Chang, F. Xu, H. Zhang, L1/2 regularization: a thresholding representation theory and a fast solver, IEEE Trans. Neural Netw. Learn. Syst. 23 (2012) 1013–1027 [doi], doi:10.1109/TNNLS.2012.2197412.
[11] Y. Wu, Elastic net for Cox's proportional hazards model with a solution path algorithm, Stat. Sin. 22 (2012) 27–294, doi:10.5705/ss.2010.107.
[12] N. Becker, G. Toedt, P. Lichter, A. Benner, Elastic SCAD as a novel penalization method for SVM classification tasks in high-dimensional data, BMC Bioinform. 12 (2011) 138 [doi], doi:10.1186/1471-2105-12-138.
[13] H. Zou, T. Hastie, Regularization and variable selection via the elastic net, J. R. Stat. Soc. Ser. B (Stat. Methodol 67 (2005) 301–320.
[14] H.-H. Huang, X.-Y. Liu, Y. Liang, Feature selection and cancer classification via sparse logistic regression with the hybrid L1/2 +2 regularization, PLoS One 11 (2016) e0149675, doi:10.1371/journal.pone.0149675.
[15] D.R. Cox, Partial likelihood, Biometrika 62 (1975) 269, doi:10.2307/2335362.
[16] K. Knight, W. Fu, Asymptotics for lasso-type estimators, Ann. Stat. (2000) 1356–1378.
[17] D. Malioutov, M. Çetin, A.S. Willsky, A sparse signal reconstruction perspective for source localization with sensor arrays, Signal Process. IEEE Trans. 53 (2005) 3010–3022.
[18] Y. Liang, C. Liu, X.Z. Luan, K.S. Leung, T.M. Chan, Z.B. Xu, H. Zhang, Sparse logistic regression with a L1/2 penalty for gene selection in cancer classification, BMC Bioinform. 14 (2013) 198, doi:10.1186/1471-2105-14-198.
[19] H.-H. Huang, Y. Liang, X.-Y. Liu, Network-based logistic classification with an enhanced L 1/2 solver reveals biomarker and subnetwork signatures for diagnosing lung cancer, Biomed. Res. Int. 2015 (2015) 713953, doi:10.1155/2015/713953.
[20] H.-H. Huang, X.-Y. Liu, Y. Liang, H. Chai, L.-Y. Xia, Identification of 13 blood-based gene expression signatures to accurately distinguish tuberculosis from other pulmonary diseases and healthy controls, Biomed. Mater. Eng. 26 (Suppl 1) (2015) S1837–S1843, doi:10.3233/BME-151486.
[21] J. Gui, H. Li, Penalized Cox regression analysis in the high-dimensional and low-sample size settings, with applications to microarray gene expression data, Bioinformatics 21 (2005) 3001–3008, doi:10.1093/bioinformatics/bti422.
[22] J. Friedman, T. Hastie, R. Tibshirani, Regularization paths for generalized linear models via coordinate descent, J. Stat. Softw. 33 (2010) 1–22.
[23] H.C. van Houwelingen, T. Bruinsma, A.A.M. Hart, L.J. van't Veer, L.F.A. Wessels, Cross-validated Cox regression on microarray gene expression data, Stat. Med. 25 (2006) 3201–3216, doi:10.1002/sim.2353.
[24] E. Graf, C. Schmoor, W. Sauerbrei, M. Schumacher, Assessment and comparison of prognostic classification schemes for survival data. *Stat. Med.* **18, 1999,** 2529–2545. http://www.ncbi.nlm.nih.gov/pubmed/10474158.
[25] P.K. Andersen, in: Frank E. Harrell (Ed.), Regression Modeling Strategies With Applications to Linear Models, Logistic Regression and Survival Analysis, Springer-Verlag, New York, 2001, p. 568. JunISBN 0-387-95232-2, *Stat. Med.* **22,** 2003, 2531–2532. doi:10.1002/sim.1497.
[26] R. Bender, T. Augustin, M. Blettner, Generating survival times to simulate Cox proportional hazards models, Stat. Med. 24 (2005) 1713–1723, doi:10.1002/sim.2059.
[27] W. Zhang, Y.W. Wan, G.I. Allen, K. Pang, M.L. Anderson, Z. Liu, Molecular pathway identification using biological network-regularized logistic models, BMC Genom. 14 (Suppl 8) (2013) S7–2164–14–S8–S7. Epub 2013 Dec 9[doi], doi:10.1186/1471-2164-14-S8-S7.
[28] T. Hastie, R. Tibshirani, J. Friedman, J. Franklin, The elements of statistical learning: data mining, inference and prediction, Math. Intell. 27 (2005) 83–85.
[29] H. Størvold, O. Lingjærde, Predicting survival from gene expression data by generalized partial least squares regression, Breast Cancer Res. 7 (2005) P4.43, doi:10.1186/bcr1173.
[30] L. Bullinger, K. Döhner, E. Bair, S. Fröhling, R.F. Schlenk, R. Tibshirani, H. Döhner, J.R. Pollack, Use of gene-expression profiling to identify prognostic subclasses in adult acute myeloid leukemia, N. Engl. J. Med. 350 (2004) 1605–1616, doi:10.1056/NEJMoa031046.
[31] P. Jézéquel, D. Loussouarn, C. Guérin-Charbonnel, L. Campion, A. Vanier, W. Gouraud, H. Lasla, C. Guette, I. Valo, V. Verrièle, M. Campone, Gene-expression molecular subtyping of triple-negative breast cancer tumours: importance of immune response, Breast Cancer Res 17 (2015) 43, doi:10.1186/s13058-015-0550-y.
[32] D.-T. Chen, A.H. Davis-Yadley, P.-Y. Huang, K. Husain, B.A. Centeno, J. Permuth-Wey, J.M. Pimiento, M. Malafa, Prognostic fifteen-gene signature for early stage pancreatic ductal adenocarcinoma, PLoS One 10 (2015) e0133562, doi:10.1371/journal.pone.0133562.
[33] A. Rosenwald, G. Wright, W.C. Chan, J.M. Connors, E. Campo, R.I. Fisher, R.D. Gascoyne, H.K. Muller-Hermelink, E.B. Smeland, J.M. Giltnane, E.M. Hurt, H. Zhao, L. Averett, L. Yang, W.H. Wilson, E.S. Jaffe, R. Simon, R.D. Klausner, J. Powell, P.L. Duffey, D.L. Longo, T.C. Greiner, D.D. Weisenburger, W.G. Sanger, B.J. Dave, J.C. Lynch, J. Vose, J.O. Armitage, E. Montserrat, A. López-Guillermo, T.M. Grogan, T.P. Miller, M. LeBlanc, G. Ott, S. Kvaloy, J. Delabie, H. Holte, P. Krajci, T. Stokke, L.M. Staudt, Lymphoma/Leukemia molecular profiling project, the use of molecular profiling to predict survival after chemotherapy for diffuse large-B-cell lymphoma, N. Engl. J. Med 346 (2002) 1937–1947, doi:10.1056/NEJMoa012914.
[34] H. Cirenajwis, H. Ekedahl, M. Lauss, K. Harbst, A. Carneiro, J. Enoksson, F. Rosengren, L. Werner-Hartman, T. Törngren, A. Kvist, E. Fredlund, P.-O. Bendahl, K. Jirström, L. Lundgren, J. Howlin, Å. Borg, S.K. Gruvberger-Saal, L.H. Saal, K. Nielsen, M. Ringnér, H. Tsao, H. Olsson, C. Ingvar, J. Staaf, G. Jönsson, Molecular stratification of metastatic melanoma using gene expression profiling : Prediction of survival outcome and benefit from molecular targeted therapy, Oncotarget 6 (2015) 12297–12309, doi:10.18632/oncotarget.3655.
[35] J. Staaf, G. Jönsson, M. Jönsson, A. Karlsson, S. Isaksson, A. Salomonsson, H.M. Pettersson, M. Soller, S.-B. Ewers, L. Johansson, P. Jönsson, M. Planck, Relation between smoking history and gene expression profiles in lung adenocarcinomas, BMC Med. Genom. 5 (2012) 22, doi:10.1186/1755-8794-5-22.
[36] Y. Liang, H. Gao, S.-Y. Lin, J.A. Goss, C. Du, K. Li, Mcph1/Brit1 deficiency promotes genomic instability and tumor formation in a mouse model, Oncogene 34 (2015) 4368–4378, doi:10.1038/onc.2014.367.

[37] D.C. Koboldt, R.S. Fulton, M.D. McLellan, H. Schmidt, J. Kalicki-Veizer, J.F. McMichael, L.L. Fulton, D.J. Dooling, L. Ding, E.R. Mardis, R.K. Wilson, A. Ally, M. Balasundaram, Y.S.N. Butterfield, R. Carlsen, C. Carter, A. Chu, E. Chuah, H.-J.E. Chun, R.J.N. Coope, N. Dhalla, R. Guin, C. Hirst, M. Hirst, R.A. Holt, D. Lee, H.I. Li, M. Mayo, R.A. Moore, A.J. Mungall, E. Pleasance, A. Gordon Robertson, J.E. Schein, A. Shafiei, P. Sipahimalani, J.R. Slobodan, D. Stoll, A. Tam, N. Thiessen, R.J. Varhol, N. Wye, T. Zeng, Y. Zhao, I. Birol, S.J.M. Jones, M.A. Marra, A.D. Cherniack, G. Saksena, R.C. Onofrio, N.H. Pho, S.L. Carter, S.E. Schumacher, B. Tabak, B. Hernandez, J. Gentry, H. Nguyen, A. Crenshaw, K. Ardlie, R. Beroukhim, W. Winckler, G. Getz, S.B. Gabriel, M. Meyerson, L. Chin, P.J. Park, R. Kucherlapati, K.A. Hoadley, J. Todd Auman, C. Fan, Y.J. Turman, Y. Shi, L. Li, M.D. Topal, X. He, H.-H. Chao, A. Prat, G.O. Silva, M.D. Iglesia, W. Zhao, J. Usary, J.S. Berg, M. Adams, J. Booker, J. Wu, A. Gulabani, T. Bodenheimer, A.P. Hoyle, J.V. Simons, M.G. Soloway, L.E. Mose, S.R. Jefferys, S. Balu, J.S. Parker, D. Neil Hayes, C.M. Perou, S. Malik, S. Mahurkar, H. Shen, D.J. Weisenberger, T. Triche Jr, P.H. Lai, M.S. Bootwalla, D.T. Maglinte, B.P. Berman, D.J. Van Den Berg, S.B. Baylin, P.W. Laird, C.J. Creighton, L.A. Donehower, G. Getz, M. Noble, D. Voet, G. Saksena, N. Gehlenborg, D. DiCara, J. Zhang, H. Zhang, C.-J. Wu, S. Yingchun Liu, M.S. Lawrence, L. Zou, A. Sivachenko, P. Lin, P. Stojanov, R. Jing, J. Cho, R. Sinha, R.W. Park, M.-D. Nazaire, J. Robinson, H. Thorvaldsdottir, J. Mesirov, P.J. Park, L. Chin, S. Reynolds, R.B. Kreisberg, B. Bernard, R. Bressler, T. Erkkila, J. Lin, V. Thorsson, W. Zhang, I. Shmulevich, G. Ciriello, N. Weinhold, N. Schultz, J. Gao, E. Cerami, B. Gross, A. Jacobsen, R. Sinha, B. Arman Aksoy, Y. Antipin, B. Reva, R. Shen, B.S. Taylor, M. Ladanyi, C. Sander, P. Anur, P.T. Spellman, Y. Lu, W. Liu, R.R.G. Verhaak, G.B. Mills, R. Akbani, N. Zhang, B.M. Broom, T.D. Casasent, C. Wakefield, A.K. Unruh, K. Baggerly, K. Coombes, J.N. Weinstein, D. Haussler, C.C. Benz, J.M. Stuart, S.C. Benz, J. Zhu, C.C. Szeto, G.K. Scott, C. Yau, E.O. Paull, D. Carlin, C. Wong, A. Sokolov, J. Thusberg, S. Mooney, S. Ng, T.C. Goldstein, K. Ellrott, M. Grifford, C. Wilks, S. Ma, B. Craft, C. Yan, Y. Hu, D. Meerzaman, J.M. Gastier-Foster, J. Bowen, N.C. Ramirez, A.D. Black, R.E. Pyatt, P. White, E.J. Zmuda, J. Frick, T.M. Lichtenberg, R. Brookens, M.M. George, M.A. Gerken, H.A. Harper, K.M. Leraas, L.J. Wise, T.R. Tabler, C. McAllister, T. Barr, M. Hart-Kothari, K. Tarvin, C. Saller, G. Sandusky, C. Mitchell, M.V. Iacocca, J. Brown, B. Rabeno, C. Czerwinski, N. Petrelli, O. Dolzhansky, M. Abramov, O. Voronina, O. Potapova, J.R. Marks, W.M. Suchorska, D. Murawa, W. Kycler, M. Ibbs, K. Korski, A. Spychała, P. Murawa, J.J. Brzeziński, H. Perz, R. Łaźniak, M. Teresiak, H. Tatka, E. Leporowska, M. Bogusz-Czerniewicz, J. Malicki, A. Mackiewicz, M. Wiznerowicz, X. Van Le, B. Kohl, N. Viet Tien, R. Thorp, N. Van Bang, H. Sussman, B. Duc Phu, R. Hajek, N. Phi Hung, T. Viet The Phuong, H. Quyet Thang, K. Zaki Khan, R. Penny, D. Mallery, E. Curley, C. Shelton, P. Yena, J.N. Ingle, F.J. Couch, W.L. Lingle, T.A. King, A. Maria Gonzalez-Angulo, G.B. Mills, M.D. Dyer, S. Liu, X. Meng, M. Patangan, F. Waldman, H. Stöppler, W. Kimryn Rathmell, L. Thorne, M. Huang, L. Boice, A. Hill, C. Morrison, C. Gaudioso, W. Bshara, K. Daily, S.C. Egea, M.D. Pegram, C. Gomez-Fernandez, R. Dhir, R. Bhargava, A. Brufsky, C.D. Shriver, J.A. Hooke, J. Leigh Campbell, R.J. Mural, H. Hu, S. Somiari, C. Larson, B. Deyarmin, L. Kvecher, A.J. Kovatich, M.J. Ellis, T.A. King, H. Hu, F.J. Couch, R.J. Mural, T. Stricker, K. White, O. Olopade, J.N. Ingle, C. Luo, Y. Chen, J.R. Marks, F. Waldman, M. Wiznerowicz, R. Bose, L.-W. Chang, A.H. Beck, A. Maria Gonzalez-Angulo, T. Pihl, M. Jensen, R. Sfeir, A. Kahn, A. Chu, P. Kothiyal, Z. Wang, E. Snyder, J. Pontius, B. Ayala, M. Backus, J. Walton, J. Baboud, D. Berton, M. Nicholls, D. Srinivasan, R. Raman, S. Girshik, P. Kigonya, S. Alonso, R. Sanbhadti, S. Barletta, D. Pot, M. Sheth, J.A. Demchok, K.R. Mills Shaw, L. Yang, G. Eley, M.L. Ferguson, R.W. Tarnuzzer, J. Zhang, L.A.L. Dillon, K. Buetow, P. Fielding, B.A. Ozenberger, M.S. Guyer, H.J. Sofia, J.D. Palchik, Comprehensive molecular portraits of human breast tumours, Nature 490 (2012) 61–70, doi:10.1038/nature11412.

[38] A. Mitra, R.A. Fillmore, B.J. Metge, M. Rajesh, Y. Xi, J. King, J. Ju, L. Pannell, L.A. Shevde, R.S. Samant, Large isoform of MRJ (DNAJB6) reduces malignant activity of breast cancer, Breast Cancer Res 10 (2008) R22, doi:10.1186/bcr1874.